Internal Phosphorus Loading in Lakes: Geochemical Pathways and Water Quality Implications (Trial Experiment)

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Load packages

```r
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```r
library(readr)
```

```
## Warning: package 'readr' was built under R version 4.5.3
```

```r
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.3

##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 4.5.3
```

```r
library(grid)
library(ggpubr)
library(ggpattern)
```

Read the CSV file

```r
data <- read.csv(
  "Data file.csv",
  check.names = FALSE,
  na.strings = c("NA", "na", "")
)
```

Creation of figures

Figure 1: Physical characteristic variation of sediment with depth

```r
# Clean column names
colnames(data) <- trimws(colnames(data))


print(colnames(data))
```

```
##  [1] "Sample No."       "Sample depth"     "Reps"             "N [%]"
##  [5] "C [%]"            "C/N  ratio"       "Soil pH"          "Soil EC / ?Scm-1"
##  [9] "NaCl IP mg/kg"    "NaCl TP mg/kg"    "NaCl Org P mg/kg" "NaBD IP mg/kg"
## [13] "NaBD TP mg/kg"    "NaBD Org P mg/kg" "NaOH IP mg/kg"    "NaOH TP mg/kg"
## [17] "NaOH Org P mg/kg" "HCl IP mg/kg"     "HCl TP mg/kg"     "HCl Org P mg/kg"
## [21] "Residue mg/kg"
```

```r
data <- data %>%
  filter(if_any(everything(), ~ !is.na(.)))

# Fill depth values downward because only first replicate row may contain depth
data <- data %>%
```

```r
  fill(`Sample depth`, .direction = "down")

# Convert depth to numeric
data <- data %>%
  mutate(`Sample depth` = as.numeric(`Sample depth`))

# ==========================================
## Summarize data by depth
# pH and EC = single available value
# N, C, C/N = mean of 3 replicates

summary_data <- data %>%
  group_by(`Sample depth`) %>%
  summarise(
    pH = first(na.omit(`Soil pH`)),
    EC = first(na.omit(`Soil EC / ?Scm-1`)),
    N_mean = mean(`N [%]`, na.rm = TRUE),
    C_mean = mean(`C [%]`, na.rm = TRUE),
    CN_mean = mean(`C/N  ratio`, na.rm = TRUE)
  ) %>%
  ungroup() %>%
  arrange(`Sample depth`)

# Check the summarized table
print(summary_data)
```

```
## # A tibble: 10 x 6
##    `Sample depth`    pH    EC N_mean C_mean CN_mean
##             <dbl> <dbl> <dbl>  <dbl>  <dbl>   <dbl>
##  1            0.5  5.42 18.0  0.322    2.01    6.25
##  2            1.5  5.52 16.9  0.305    1.82    5.99
##  3            2.5  5.47 15.1  0.126    1.48   11.8
##  4            3.5  5.63 14.4  0.122    1.37   11.3
##  5            4.5  5.6   9.87 0.116    1.34   11.6
##  6            6.5  5.58  7.49 0.103    1.22   11.9
##  7            9.5  5.64  7.36 0.101    1.20   11.8
##  8           12.5  5.62  5.39 0.103    1.26   12.2
##  9           15.5  5.49  2.35 0.105    1.38   13.1
## 10           19.5  5.68  2.33 0.0984   2.63   26.7
```

```r
str(summary_data)
```

```
## tibble [10 x 6] (S3: tbl_df/tbl/data.frame)
##  $ Sample depth: num [1:10] 0.5 1.5 2.5 3.5 4.5 6.5 9.5 12.5 15.5 19.5
##  $ pH          : num [1:10] 5.42 5.52 5.47 5.63 5.6 5.58 5.64 5.62 5.49 5.68
##  $ EC          : num [1:10] 18.03 16.9 15.13 14.4 9.87 ...
##  $ N_mean      : num [1:10] 0.322 0.305 0.126 0.122 0.116 ...
##  $ C_mean      : num [1:10] 2.01 1.82 1.48 1.37 1.34 ...
##  $ CN_mean     : num [1:10] 6.25 5.99 11.77 11.25 11.6 ...
```

```r
# Common theme
```

```r
my_theme <- theme_classic() +
  theme(
    axis.title = element_text(size = 8, color = "black"),
    axis.text  = element_text(size = 8, color = "black"),

    plot.title = element_text(size = 10, face = "bold",
                              hjust = 0.5, lineheight = 0.95),

    axis.line = element_line(color = "black", linewidth = 0.7),

    panel.border = element_rect(color = "black", fill = NA, linewidth = 0.8),

    plot.margin = margin(5, 5, 5, 5)
  )

# Common reversed depth axis
y_depth <- scale_y_reverse(
  limits = c(20, 0),
  breaks = seq(0, 20, 2),
  expand = c(0, 0)
)

# ================================
# a) pH variation with depth
# ==============================
p1 <- ggplot(summary_data, aes(x = pH, y = `Sample depth`)) +
  geom_path(color = "black", linewidth = 0.7) +
  geom_point(color = "black", size = 1.2) +
  y_depth +
  scale_x_continuous(
    breaks = c(5, 5.5, 6),
    limits = c(5, 6)
  ) +
  labs(
    title = "a) pH variation with depth\n",
    x = "pH",
    y = "Sediment Depth (cm)"
  ) +
  my_theme

# ===========================
# b) Electrical Conductivity variation with depth
# ===========================
p2 <- ggplot(summary_data, aes(x = EC, y = `Sample depth`)) +
  geom_path(color = "black", linewidth = 0.7) +
  geom_point(color = "black", size = 1.2) +
  y_depth +
  labs(
    title = "b) Electrical Conductivity\nvariation with depth",
    x = expression(paste("Electrical Conductivity (", mu, "S ", cm^-1, ")")),
    y = "Sediment Depth (cm)"
  ) +
  my_theme
```

```r
# ====================================
# c) Nitrogen percentage variation with depth
# ====================================
p3 <- ggplot(summary_data, aes(x = N_mean, y = `Sample depth`)) +
  geom_path(color = "black", linewidth = 0.7) +
  geom_point(color = "black", size = 1.2) +
  y_depth +
  labs(
    title = "c) Nitrogen percentage\nvariation with depth",
    x = "Nitrogen percentage (%)",
    y = "Sediment Depth (cm)"
  ) +
  my_theme

# ==========================================
# d) Carbon percentage variation with depth
# ==========================================
p4 <- ggplot(summary_data, aes(x = C_mean, y = `Sample depth`)) +
  geom_path(color = "black", linewidth = 0.7) +
  geom_point(color = "black", size = 1.2) +
  y_depth +
  labs(
    title = "d) Carbon percentage\nvariation with depth",
    x = "Carbon percentage (%)",
    y = "Sediment Depth (cm)"
  ) +
  my_theme

# ==========================================
# e) C/N variation with depth
# ==========================================
p5 <- ggplot(summary_data, aes(x = CN_mean, y = `Sample depth`)) +
  geom_path(color = "black", linewidth = 0.7) +
  geom_point(color = "black", size = 1.2) +
  y_depth +
  labs(
    title = "e) C/N variation with depth\n",
    x = "C/N ratio",
    y = "Sediment Depth (cm)"
  ) +
  my_theme

# ==========================================
# Combine all five plots
# ==========================================
combined_plot <- ggarrange(
  p1, p2, p3, p4, p5,
  ncol = 3, nrow = 2
)
```

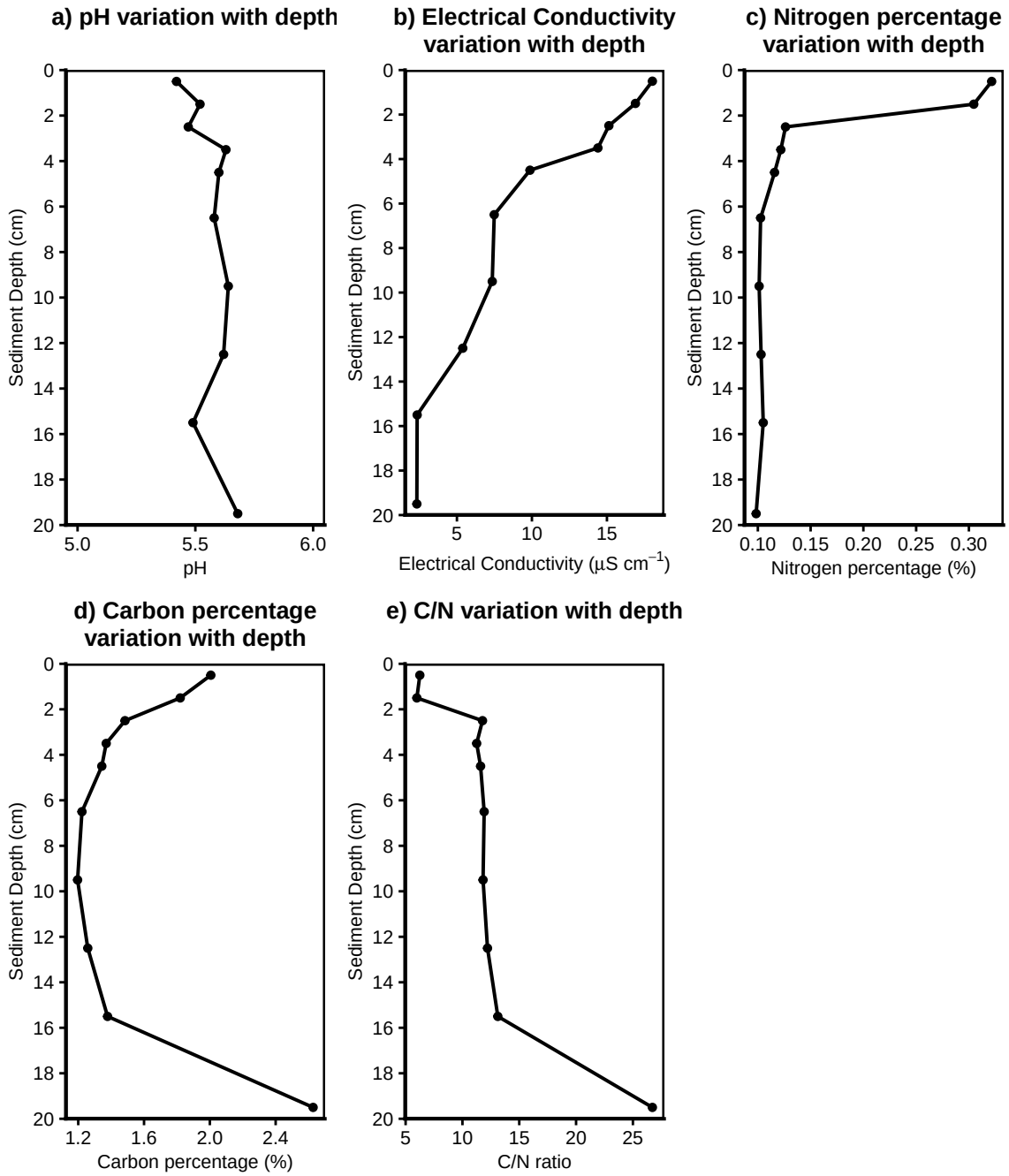


Figure 2: Distribution of Pi and Po with depth

```
# -------------------------
# 1. Read and clean CSV
# -------------------------
data <- read.csv(
  "Data file.csv",
  check.names = FALSE,
  na.strings = c("NA", "na", "")
```

```r
)

names(data) <- trimws(names(data))

# Rename only the needed columns to match the original code
data <- data %>%
  rename(
    `Sample depth` = `Sample depth`,
    `Loosely sorbed Pi (mg kg-1)` = `NaCl IP mg/kg`,
    `Loosely sorbed Po (mg kg-1)` = `NaCl Org P mg/kg`,
    `Loosely sorbed Pt (mg kg-1)` = `NaCl TP mg/kg`,
    `Redox sensitive Pi (mg kg-1)` = `NaBD IP mg/kg`,
    `Redox sensitive Po (mg kg-1)` = `NaBD Org P mg/kg`,
    `Redox sensitive Pt (mg kg-1)` = `NaBD TP mg/kg`,
    `Fe/Al Oxide Pi (mg kg-1)` = `NaOH IP mg/kg`,
    `Fe/Al Oxide Po (mg kg-1)` = `NaOH Org P mg/kg`,
    `Fe/Al Oxide Pt (mg kg-1)` = `NaOH TP mg/kg`,
    `Apatite bound Pi (mg kg-1)` = `HCl IP mg/kg`,
    `Apatite bound Po (mg kg-1)` = `HCl Org P mg/kg`,
    `Apatite bound Pt (mg kg-1)` = `HCl TP mg/kg`,
    `Residual P (mg kg-1)` = `Residue mg/kg`
  )

data <- data %>%
  fill(`Sample depth`, .direction = "down")

numeric_cols <- c(
  "Sample depth",
  "Loosely sorbed Pi (mg kg-1)",
  "Loosely sorbed Po (mg kg-1)",
  "Loosely sorbed Pt (mg kg-1)",
  "Redox sensitive Pi (mg kg-1)",
  "Redox sensitive Po (mg kg-1)",
  "Redox sensitive Pt (mg kg-1)",
  "Fe/Al Oxide Pi (mg kg-1)",
  "Fe/Al Oxide Po (mg kg-1)",
  "Fe/Al Oxide Pt (mg kg-1)",
  "Apatite bound Pi (mg kg-1)",
  "Apatite bound Po (mg kg-1)",
  "Apatite bound Pt (mg kg-1)",
  "Residual P (mg kg-1)"
)

data[numeric_cols] <- lapply(data[numeric_cols], as.numeric)


# 2. Summary function
# Pt_mean = mean(Pi) + mean(Po)
# --------------------------
make_fraction_summary <- function(data_in, pi_col, po_col, fraction_name) {

  tmp <- data_in %>%
    transmute(
```

```r
      `Sample depth`,
      Pi_value = .data[[pi_col]],
      Po_value = .data[[po_col]],
      Pt_rep   = Pi_value + Po_value
    )

  out <- tmp %>%
    group_by(`Sample depth`) %>%
    summarise(
      Pi_mean = mean(Pi_value, na.rm = TRUE),
      Po_mean = mean(Po_value, na.rm = TRUE),
      Pt_rep_sd = sd(Pt_rep, na.rm = TRUE),
      n_pt = sum(!is.na(Pt_rep)),
      .groups = "drop"
    ) %>%
    mutate(
      Pt_mean = Pi_mean + Po_mean,
      Pt_se = Pt_rep_sd / sqrt(n_pt)
    ) %>%
    arrange(`Sample depth`) %>%
    mutate(
      Fraction = fraction_name,
      y = rev(seq_len(n()))
    )

  return(out)
}

# 3. Plot function
# -------------------------
plot_fraction <- function(summary_df, title_text, dark_col, light_col,
                          x_max, x_breaks) {

  bar_h <- 0.75

  pi_rect <- summary_df %>%
    mutate(
      xmin = 0,
      xmax = Pi_mean,
      ymin = y - bar_h / 2,
      ymax = y + bar_h / 2
    )

  po_rect <- summary_df %>%
    mutate(
      xmin = Pi_mean,
      xmax = Pi_mean + Po_mean,
      ymin = y - bar_h / 2,
      ymax = y + bar_h / 2
    )

  pt_rect <- summary_df %>%
    mutate(
```

```r
    xmin = 0,
    xmax = Pt_mean,
    ymin = y - bar_h / 2,
    ymax = y + bar_h / 2
  )

legend_df <- data.frame(
  x = c(1, 1, 1),
  y = c(1, 1, 1),
  grp = factor(c("Pt", "Pi", "Po"), levels = c("Pt", "Pi", "Po"))
)

ggplot() +
  geom_rect(
    data = pi_rect,
    aes(xmin = xmin, xmax = xmax, ymin = ymin, ymax = ymax),
    fill = dark_col,
    color = NA
  ) +
  geom_rect(
    data = po_rect,
    aes(xmin = xmin, xmax = xmax, ymin = ymin, ymax = ymax),
    fill = light_col,
    color = NA
  ) +
  geom_rect(
    data = pt_rect,
    aes(xmin = xmin, xmax = xmax, ymin = ymin, ymax = ymax),
    fill = NA,
    color = "black",
    linewidth = 0.8
  ) +
  geom_errorbarh(
    data = summary_df,
    aes(
      y = y,
      xmin = pmax(Pt_mean - Pt_se, 0),
      xmax = Pt_mean + Pt_se
    ),
    height = 0.18,
    linewidth = 0.8,
    color = "black"
  ) +
  geom_point(
    data = legend_df,
    aes(x = x, y = y, fill = grp, shape = grp),
    size = 5,
    color = "black",
    stroke = 1,
    alpha = 0
  ) +
  scale_y_continuous(
    breaks = summary_df$y,
```

```r
      labels = summary_df$`Sample depth`,
      expand = expansion(mult = c(0.02, 0.02))
    ) +
    scale_x_continuous(
      limits = c(0, x_max),
      breaks = x_breaks,
      expand = c(0, 0.05)
    ) +
    scale_fill_manual(
      values = c(
        "Pt" = "white",
        "Pi" = dark_col,
        "Po" = light_col
      ),
      breaks = c("Pt", "Pi", "Po"),
      labels = c("Pt", "Pi", "Po"),
      name = NULL,
      guide = guide_legend(
        override.aes = list(
          shape = c(22, 22, 22),
          size = c(6, 6, 6),
          alpha = c(1, 1, 1),
          color = c("black", NA, NA),
          fill = c("white", dark_col, light_col),
          stroke = c(1.2, 0, 0)
        )
      )
    ) +
    scale_shape_manual(
      values = c("Pt" = 22, "Pi" = 22, "Po" = 22),
      guide = "none"
    ) +
    labs(
      title = title_text,
      x = expression(PO[4]^{"3-"}~"- P concentration (mg kg"^{-1}*")"),
      y = "Sediment Depth (cm)"
    ) +
    theme_classic() +
    theme(
      plot.title = element_text(size = 12, hjust = 0.5),
      axis.title.x = element_text(size = 10),
      axis.title.y = element_text(size = 10),
      axis.text.x = element_text(size = 8, color = "black"),
      axis.text.y = element_text(size = 8, color = "black"),
      legend.text = element_text(size = 8),
      legend.position = c(0.78, 0.4),
      legend.key.size = unit(0.2, "cm"),
      panel.border = element_rect(color = "black", fill = NA, linewidth = 0.6),
      axis.line = element_blank(),
      plot.margin = margin(6, 10, 6, 10)
    )
}
```

```r
# 4. Color selection
# -------------------------
purple_dark  <- "#7B1FA2"
purple_light <- "#C5ADD8"

red_dark     <- "#E52424"
red_light    <- "#DDA3A3"

green_dark   <- "#329632"
green_light  <- "#A8C6A8"

blue_dark    <- "#3774C2"
blue_light   <- "#9EB7D3"

xmax_loose   <- 20
breaks_loose <- seq(0, 20, by = 5)

xmax_redox   <- 150
breaks_redox <- seq(0, 150, by = 50)

xmax_feal    <- 900
breaks_feal  <- seq(0, 900, by = 150)

xmax_apatite <- 100
breaks_apatite <- seq(0, 100, by = 20)


# 5. Specs for loop
# -------------------------
fraction_specs <- list(
  list(
    title = "a) Loosely sorbed P",
    pi_col = "Loosely sorbed Pi (mg kg-1)",
    po_col = "Loosely sorbed Po (mg kg-1)",
    frac_name = "Loosely sorbed P",
    dark = purple_dark,
    light = purple_light,
    xmax = xmax_loose,
    breaks = breaks_loose
  ),
  list(
    title = "b) Redox sensitive P",
    pi_col = "Redox sensitive Pi (mg kg-1)",
    po_col = "Redox sensitive Po (mg kg-1)",
    frac_name = "Redox sensitive P",
    dark = green_dark,
    light = green_light,
    xmax = xmax_redox,
    breaks = breaks_redox
  ),
  list(
    title = "c) Fe/Al Oxide P",
    pi_col = "Fe/Al Oxide Pi (mg kg-1)",
```

```r
    po_col = "Fe/Al Oxide Po (mg kg-1)",
    frac_name = "Fe/Al Oxide P",
    dark = red_dark,
    light = red_light,
    xmax = xmax_feal,
    breaks = breaks_feal
  ),
  list(
    title = "d) Apatite bound P",
    pi_col = "Apatite bound Pi (mg kg-1)",
    po_col = "Apatite bound Po (mg kg-1)",
    frac_name = "Apatite bound P",
    dark = blue_dark,
    light = blue_light,
    xmax = xmax_apatite,
    breaks = breaks_apatite
  )
)


# 6. Create plots with loop
# -------------------------
plot_list <- vector("list", length(fraction_specs))

for (i in seq_along(fraction_specs)) {
  spec <- fraction_specs[[i]]

  summary_df <- make_fraction_summary(
    data_in = data,
    pi_col = spec$pi_col,
    po_col = spec$po_col,
    fraction_name = spec$frac_name
  )

  plot_list[[i]] <- plot_fraction(
    summary_df = summary_df,
    title_text = spec$title,
    dark_col = spec$dark,
    light_col = spec$light,
    x_max = spec$xmax,
    x_breaks = spec$breaks
  )
}
```

```
## Warning: `geom_errorbarh()` was deprecated in ggplot2 4.0.0.
## i Please use the `orientation` argument of `geom_errorbar()` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```r
# keep same object names if you want to use them later
p1 <- plot_list[[1]]
p2 <- plot_list[[2]]
```

```r
p3 <- plot_list[[3]]
p4 <- plot_list[[4]]


# 7. Combine
# -------------------------
final_plot <- ggarrange(
  plotlist = plot_list,
  ncol = 2, nrow = 2
)
```

```
## 'height' was translated to 'width'.
## 'height' was translated to 'width'.
## 'height' was translated to 'width'.
## 'height' was translated to 'width'.
```

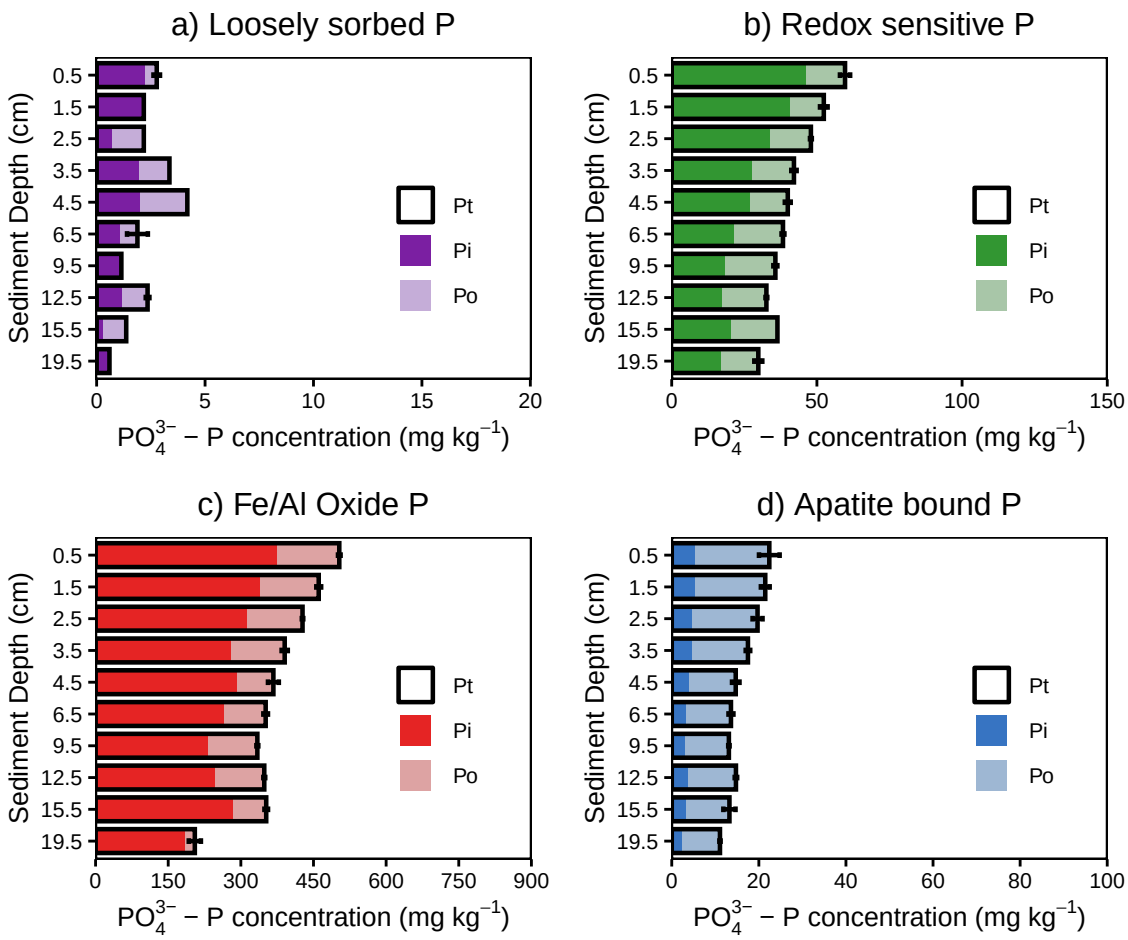


Figure 3: Percent distribution of P fractions with depth

```r
# 1. Read and clean CSV
# -------------------------
```

```r
data <- read.csv(
  "Data file.csv",
  check.names = FALSE,
  na.strings = c("NA", "na", "")
)

names(data) <- trimws(names(data))

# 2. Rename Columns
data <- data %>%
  rename(
    sample_no    = `Sample No.`,
    sample_depth = `Sample depth`,
    rep          = `Reps`,
    ph           = `Soil pH`,
    ec           = `Soil EC / ?Scm-1`,
    nacl_ip      = `NaCl IP mg/kg`,
    nacl_tp      = `NaCl TP mg/kg`,
    nacl_org     = `NaCl Org P mg/kg`,
    nabd_ip      = `NaBD IP mg/kg`,
    nabd_tp      = `NaBD TP mg/kg`,
    nabd_org     = `NaBD Org P mg/kg`,
    naoh_ip      = `NaOH IP mg/kg`,
    naoh_tp      = `NaOH TP mg/kg`,
    naoh_org     = `NaOH Org P mg/kg`,
    hcl_ip       = `HCl IP mg/kg`,
    hcl_tp       = `HCl TP mg/kg`,
    hcl_org      = `HCl Org P mg/kg`,
    residue      = `Residue mg/kg`
  )

# 3. Clean and Process Data
data_clean <- data %>%
  mutate(across(c(sample_no, sample_depth, ph, ec, contains("_"), residue), as.numeric)) %>%
  mutate(rep = as.character(rep)) %>%
  fill(sample_depth, .direction = "down")

# 4. Calculate Percentages
pct_rep <- data_clean %>%
  transmute(
    depth   = sample_depth,
    rep     = rep,
    nacl_tp = replace_na(nacl_tp, 0),
    nabd_tp = replace_na(nabd_tp, 0),
    naoh_tp = replace_na(naoh_tp, 0),
    hcl_tp  = replace_na(hcl_tp, 0),
    residue = replace_na(residue, 0)
  ) %>%
  mutate(
    total_p     = nacl_tp + nabd_tp + naoh_tp + hcl_tp + residue,
    nacl_pct    = (nacl_tp / total_p) * 100,
    nabd_pct    = (nabd_tp / total_p) * 100,
    naoh_pct    = (naoh_tp / total_p) * 100,
```

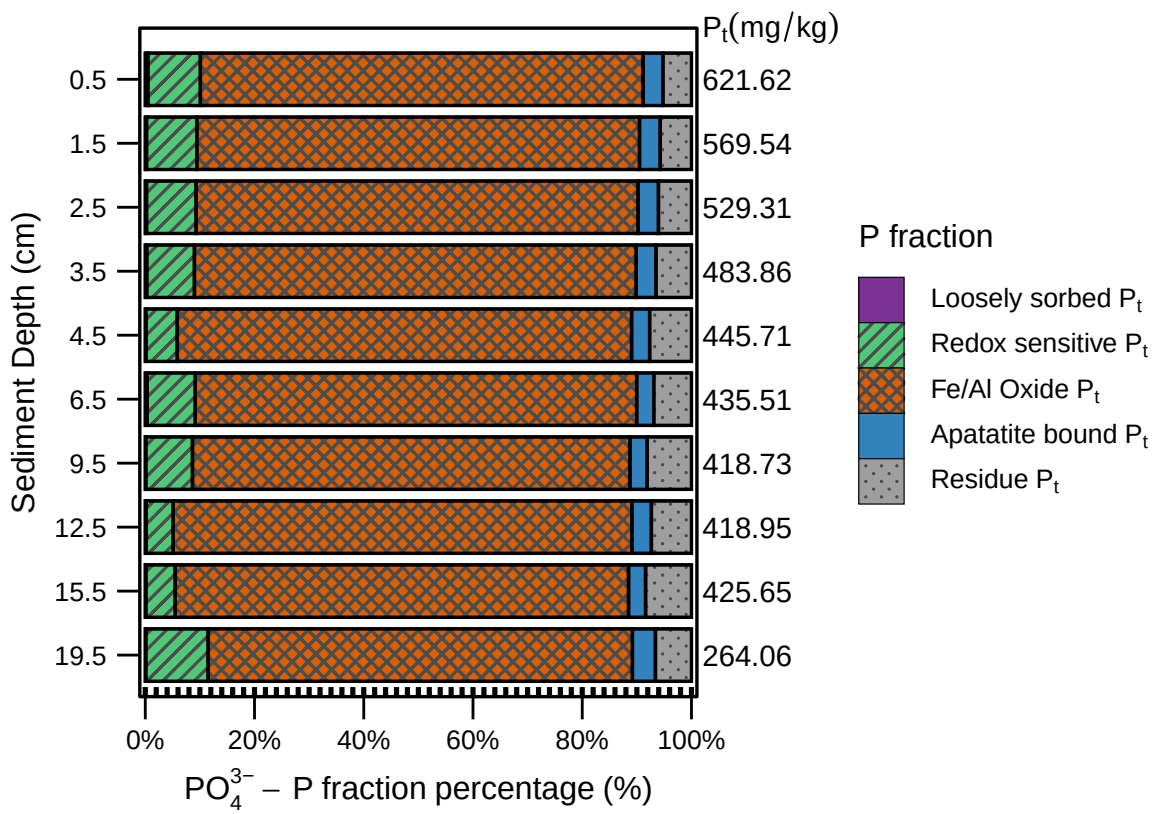
```r
    hcl_pct     = (hcl_tp / total_p) * 100,
    residue_pct = (residue / total_p) * 100
  )

# 5. Renaming panel titles
pct_long <- pct_rep %>%
  select(depth, rep, contains("_pct")) %>%
  pivot_longer(
    cols      = contains("_pct"),
    names_to  = "fraction",
    values_to = "percent"
  ) %>%
  mutate(
    fraction = factor(
      fraction,
      levels = c("nacl_pct", "nabd_pct", "naoh_pct", "hcl_pct", "residue_pct"),
      labels = c(
        "Loosely sorbed Pt",
        "Redox sensitive Pt",
        "Fe/Al Oxide Pt",
        "Apatatite bound Pt",
        "Residue Pt"
      )
    )
  )

# 6. Summary Statistics
sum_df <- pct_long %>%
  group_by(depth, fraction) %>%
  summarise(
    mean_percent = mean(percent, na.rm = TRUE),
    se_percent   = sd(percent, na.rm = TRUE) / sqrt(sum(!is.na(percent))),
    .groups      = "drop"
  ) %>%
  group_by(depth) %>%
  arrange(fraction, .by_group = TRUE) %>%
  mutate(xmax = cumsum(mean_percent)) %>%
  ungroup()

# 6A. Calculate Total P (mean across replicates)
totalP_df <- data_clean %>%
  transmute(
    depth   = sample_depth,
    total_p = rowSums(across(c(nacl_tp, nabd_tp, naoh_tp, hcl_tp, residue)), na.rm = TRUE)
  ) %>%
  group_by(depth) %>%
  summarise(
    total_p_mean = mean(total_p, na.rm = TRUE),
    .groups      = "drop"
  )

# 7. Formatting Depth Labels
depth_order <- sort(unique(sum_df$depth))
```

```r
depth_key <- data.frame(
  depth = depth_order,
  depth_label = as.character(depth_order)
)

sum_df <- sum_df %>%
  left_join(depth_key, by = "depth") %>%
  mutate(depth_f = factor(depth_label, levels = rev(depth_key$depth_label))) %>%
  left_join(totalP_df, by = "depth")

outline_df <- sum_df %>%
  distinct(depth_f) %>%
  mutate(total = 100)

# 7A. Labels for Total P
label_df <- sum_df %>%
  distinct(depth_f, total_p_mean) %>%
  mutate(
    x_label = 102,
    label   = sprintf("%.2f", total_p_mean)
  )

# 8. Colors on plot
fill_cols <- c(
  "Loosely sorbed Pt"   = "#7B3294",
  "Redox sensitive Pt"  = "#50C878",
  "Fe/Al Oxide Pt"      = "#d95f02",
  "Apatatite bound Pt"  = "#3182BD",
  "Residue Pt"          = "#9E9E9E"
)

pattern_vals <- c(
  "Loosely sorbed Pt"   = "none",
  "Redox sensitive Pt"  = "stripe",
  "Fe/Al Oxide Pt"      = "crosshatch",
  "Apatatite bound Pt"  = "none",
  "Residue Pt"          = "circle"
)

pattern_angle_vals <- c(
  "Loosely sorbed Pt"   = 0,
  "Redox sensitive Pt"  = 45,
  "Fe/Al Oxide Pt"      = 45,
  "Apatatite bound Pt"  = 90,
  "Residue Pt"          = 45
)

# 9. Theme and guide settings
base_theme <- theme_bw(base_size = 20) +
  theme(
    panel.grid       = element_blank(),
    axis.title.x     = element_text(size = 12),
    axis.title.y     = element_text(size = 12),
```

```r
    axis.text.x      = element_text(size = 10),
    axis.text.y      = element_text(size = 10),
    axis.ticks       = element_line(colour = "black", linewidth = 0.5),
    axis.ticks.length = unit(0.3, "cm"),
    axis.text        = element_text(colour = "black"),
    legend.title     = element_text(size = 12, colour = "black"),
    legend.text      = element_text(size = 10),
    legend.key.size  = unit(0.6, "cm"),
    legend.key.height = unit(0.6, "cm"),
    legend.key.width = unit(0.6, "cm"),
    legend.spacing.y = unit(0.1, "cm"),
    legend.position  = c(1.55, 0.5),
    panel.border     = element_rect(colour = "black", fill = NA, linewidth = 0.6),
    plot.margin      = unit(c(0.5, 6, 0.5, 0.5), "cm"),
    aspect.ratio     = 1.2
  )

legend_override <- guide_legend(
  override.aes = list(
    pattern         = unname(pattern_vals),
    pattern_angle   = unname(pattern_angle_vals),
    pattern_fill    = rep("grey30", length(pattern_vals)),
    pattern_colour  = rep("grey30", length(pattern_vals)),
    pattern_density = rep(0.03, length(pattern_vals)),
    pattern_spacing = rep(0.02, length(pattern_vals)),
    colour          = rep("black", length(pattern_vals)),
    linewidth       = rep(0.6, length(pattern_vals))
  )
)

# 10. Create Plot
p <- ggplot(
  sum_df,
  aes(
    y = depth_f,
    x = mean_percent,
    fill = fraction,
    pattern = fraction,
    pattern_angle = fraction
  )
) +
  geom_col_pattern(
    position = position_stack(reverse = TRUE),
    colour = "black",
    linewidth = 0.6,
    width = 0.82,
    pattern_fill = "grey30",
    pattern_colour = "grey30",
    pattern_density = 0.03,
    pattern_spacing = 0.02,
    pattern_res = 600,
    pattern_key_scale_factor = 0.7
  ) +
```

```r
  geom_col(
    data = outline_df,
    aes(x = total, y = depth_f),
    inherit.aes = FALSE,
    fill = NA,
    colour = "black",
    linewidth = 0.5,
    width = 0.82
  ) +
  annotate(
    "segment",
    x = seq(0, 100, 2),
    xend = seq(0, 100, 2),
    y = 0.5,
    yend = 0.35,
    colour = "black"
  ) +
  geom_text(
    data = label_df,
    aes(x = x_label, y = depth_f, label = label),
    inherit.aes = FALSE,
    hjust = 0,
    size = 4
  ) +
  annotate(
    "text",
    x = 102,
    y = length(unique(sum_df$depth_f)) + 0.8,
    label = "P[t] (mg/kg)",
    parse = TRUE,
    hjust = 0,
    size = 4
  ) +
  scale_fill_manual(
    values = fill_cols,
    name = "P fraction",
    labels = c(
      expression("Loosely sorbed " * P[t]),
      expression("Redox sensitive " * P[t]),
      expression("Fe/Al Oxide " * P[t]),
      expression("Apatatite bound " * P[t]),
      expression("Residue " * P[t])
    ),
    guide = legend_override
  ) +
  scale_pattern_manual(
    values = pattern_vals,
    guide = "none"
  ) +
  scale_pattern_angle_manual(
    values = pattern_angle_vals,
    guide = "none"
  ) +
```

```r
  scale_x_continuous(
    breaks = seq(0, 100, 20),
    labels = function(x) paste0(x, "%"),
    expand = c(0.01, 0)
  ) +
  coord_cartesian(xlim = c(0, 100), clip = "off") +
  labs(
    x = expression(PO[4]^{"3-"} ~ - ~ P ~ fraction ~ percentage ~ "(%)"),
    y = "Sediment Depth (cm)"
  ) +
  base_theme

print(p)
```



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